

DSCI 320 Project Proposal

Exploring BFGS for training neural networks on the Spotify Music Dataset

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Project Proposal Description

We propose to formulate an optimization problem to predict the popularity of a song based on its audio features. These features are prescribed via the Spotify dataset that categorizes a song by rating it over a few different characteristics. If we can determine what features correlate most with popularity, we can determine which songs will be recommended more often.

Dataset

Our project will use the [Spotify Music Dataset](#). The dataset contains track-level audio features (e.g., danceability, energy, tempo, loudness) along with genre and popularity indicators.

Model

We will be using ridge regression as our primary model. Ridge regression is effective on high dimensional data sets such as the Spotify dataset since each song has several audio features that could have correlation.

Algorithms

To support our ridge regression model, we will be using the following algorithms:

- Steepest Decent
- BFGS

Tentative Mathematical Topic

We will focus on the following mathematical analysis:

- Properties of convex quadratic objectives
- Convergence rates of steepest descent vs BFGS

Preliminary Outline of Comparative Analysis

With the use of Ridge Regression, we will compare the use of Steepest Descent and BFGS by looking at:

- Convergence Speed
 - We can look at how many iterations they use and see how quickly the algorithms decrease the model's error.
- Efficiency
 - We can notice which is more expensive and the overall time it takes to go through the steps of each algorithm
- Stability
 - Sensitivity on choosing a step size
- Accuracy

- Compare the final solutions to see if they are the same or as expected